

An Introduction to Computer Circuit:

In theoretical computer science, a **circuit** is a model of computation in which input values proceed through a sequence of gates, each of which computes a function. Circuits of this kind provide a generalization of Boolean circuits and a mathematical model for digital logic circuits. Circuits are defined by the gates they contain and the values the gates can produce. For example, the values in a Boolean circuit are Boolean values, and the circuit includes conjunction, disjunction, and negation gates.

Circuits for a computer, is a complete path, or a network of interconnected paths for electron flow. The concept of a computer circuit is binary, with only two possible states. They use transistors, which are electronically opened and closed in nanoseconds and picoseconds (billionths and trillionths of a second), to act as on-off switches.

The circuitry of a computer's internal workings determines how quickly it operates. Shortening the time it takes to open and close switches and creating circuit routes that can handle the higher speeds are two ways to attain faster rates.

Motherboard as the main circuit board in a PC.

A motherboard is the main printed circuit board (PCB) in a computer. The motherboard is a computer's central communications backbone connectivity point, through which all components and external peripherals connect.

Motherboards can be found in virtually all computers, especially desktop and laptop PCs. The components that connect through them include chipsets, central processing units (CPU) and memory. The external peripherals include Wi-Fi, Ethernet and graphics cards with the graphics processing unit, or GPU.

IC as the major Circuitry Technology in Modern PC.

Integrated circuits, also known as microchips, play a crucial role in modern electronics. These tiny devices can contain hundreds of thousands of transistors, resistors, and other components, allowing them to perform complex tasks in a compact and efficient manner.

There are many different types of integrated circuits available, each designed to meet specific needs and requirements. From memory chips used in computers to

microcontrollers found in automobiles, integrated circuits come in a wide range of shapes, sizes, and specifications.

An integrated circuit generally comprises parts like transistors, resistors and condensers. It has multiple terminals for different types of functions to perform, especially embedded systems. These integrated circuits are miniature and can be formed on a silicon wafer.

What is an Integrated Circuit?

An integrated circuit can be defined as a circuit where all or some circuit elements are closely and inseparably connected with each other, where it acts as a single unit for multi-functional purposes.

An integrated circuit has multiple sets of electronic components on a single flat piece which is made up of semiconductor material. MOSFETs (Metal Oxide Semiconductor Field Effect Transistors) are integrated into a very small chip. An integrated circuit is faster, more reliable, more efficient and less expensive.

Based on the construction, an integrated circuit is divided into three parts:

- a) Digital Integrated Circuit
- b) Analog Integrated Circuit
- c) Mixed Integrated Circuit